Depth-First Search: Path Finding

Design and Analysis of Algorithms

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Depth-First Search (DFS)

- ```
  DFS-Visit(i):
  Status[i] = ‘visited’
  For all j such that (i,j) is an edge:
      If Status[j] = ‘unvisited’:
          Pred[j] = i
          DFS-Visit(j)
  ```

- To find a path from i to j, launch DFS at i, then walk backward along pred[] pointers from j.
- Collectively, pred[] pointers form a directed tree rooted at i.
- Paths recovered in the same way in most shortest path algorithms, and also in dynamic programming.
- Let’s see an example in code…
  ```